

Program3: code a Dockerized python Flask or [node.js](#) application

We have to create a single folder called program 3 so that in that we can create three files called

1. Dockerfile
2. app.py
- 3.requirements.txt

```
1RV24MC036_gayatri_v_k@ubuntu:~/program2$ cd ..
1RV24MC036_gayatri_v_k@ubuntu:~$ mkdir program3
1RV24MC036_gayatri_v_k@ubuntu:~$ cd program3
1RV24MC036_gayatri_v_k@ubuntu:~/program3$ nano Dockerfile
1RV24MC036_gayatri_v_k@ubuntu:~/program3$ nano app.py
1RV24MC036_gayatri_v_k@ubuntu:~/program3$ nano requirements.txt
1RV24MC036_gayatri_v_k@ubuntu:~/program3$ nano requirements.txt
```

In nano Dockerfile we have to add the code below

```
# Use Python image
FROM python:latest
# Set working directory
WORKDIR /app
# Copy requirements and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
# Copy application code
COPY . .
# Expose port 5000
EXPOSE 5000
# run the app
CMD ["python", "app.py"]
```

In nano [app.py](#) we have to add the code below

```
from flask import Flask
app = Flask(__name__)
@app.route("/")
def home():
    return "Hello from Simple Flask Docker!"
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

In nano requirements.txt we have to add the code below

```
GNU nano 7.2  
Flask==2.3.3
```

Now have to run the following commands for the output

```
>docker build -t program3 .
```

```
> docker run -p 5000:5000 program3
```



🔒 Not Secure http://172.18.0.3:5000

Hello from Simple Flask Docker!